सहायक प्राध्यापक (प्राणिशास्त्र) महाराष्ट्र शिक्षण सेवा, गट-अ
(महाविद्यालयीन शाखा)
उच्च व तंत्र शिक्षण विभाग
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Higher and Technical Education Department

परीक्षेचे टप्पे: **१)** लेखी परीक्षा - २०० गुण **२)** मुलाखत- ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विषयांशी संबंधीत घटक (११७०)	इंग्रजी	१००	२००	एक तास	पदव्युत्तर पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विषयांशी संबंधित घटक यामध्ये खालील घटक व उपघटकांचा समावेश असेल.

अ.क्र.	घटक व उपघटक
1	**UNIT-I**
	Non-Chordata and Chordata: The arrangement and connections between different phyla up to subclasses, such as Radiata and Bilateria, Protostomes and Deuterostomes, and Acoelomate and Coelomate; The symmetrical status of Protista, Parazoa, Onychophora, and Hemichordata. a. Protozoa: General characteristics and life cycle of Paramaecium, Monocystis; locomotion, feeding, reproduction, sex. Leishmania and Plasmodium. b. Reproduction, canal system, and skeleton of Porifera. c. Cnidaria: general characteristics, life history of Obelia and Aurelia, coral reefs and their creation, polymorphism, defensive structures and their mechanism. d. Platyhelminthes: Parasitic adaptability; general characteristics, life cycle, and pathogenic signs of Taenia and Fasciola e. Nemathelminthes: Overview, life cycle, and Ascaris and Wuchereria's adaptability as parasites. f. Coelom and metamerism; polychaete life modes; general characteristics and life cycle of Nereis, earthworm, and leach. g. Arthropoda: Mould modification, parts in insects (cockroach, mosquito, housefly, honey bee, and butterfly), metamorphosis in insects and its hormonal regulation, social behavior of Apis and termites, larval forms and parasitism in Crustacea, vision and respiration in arthropods (Prawn, cockroach, and scorpion). h. Molluscs: The feeding, breathing, moving, general characteristics, and life cycle of Lamellidens, Pila, and Sepia. Gastropod torsion and detorsion. i. Echinodermata: Asterias' larval morphologies, general characteristics, breathing, feeding, and movement. j. (Protochordata: General characteristics and life cycle of Branchiostoma and Herdmania; origin of chordates.

	k. Pisces: Migration, respiration, and movement. l. Amphibians: Paedomorphosis, parental care, and the genesis of tetrapods. m. Reptilia; genesis of reptiles, varieties of skulls, Sphenodon and crocodile status. n. Aves: Migration, flight adaption, and the origin of birds. o. Mammalia: The origin of mammals, their dental structure, the basic characteristics of pouched, aquatic, and egg-laying mammals, as well as their relationships with the endocrine glands (pituitary, thyroid, parathyroid, adrenal, pancreatic, and gonads). p. A comparison of the functional anatomy of different vertebrate systems. The endoskeleton, digestive system, respiratory system, circulatory system (including the heart and aortic arches), urinogenital system, brain, and sense organs (eye and ear) are among the organs that comprise the body.
2	**UNIT- II**
	Ecological Principles **A. The Environment:** Physical environment; biotic environment; biotic and abiotic interactions. **B. Habitat and niche:** Concept of habitat and niche; niche width and overlap; fundamental and realized niche; resource partitioning; characte displacement. **C. Population ecology:** Characteristics of a population; population growth curves; population regulation; life history strategies (r and K selection); concept of metapopulation – demes and dispersal, interdemic extinctions, age structured populations. **D. Species interactions:** Types of interactions, interspecific competition, herbivory, carnivory, pollination, symbiosis. **E. Community ecology:** Nature of communities; community structure and attributes; levels of species diversity and its measurement; edges and ecotones. **F. Ecological succession:** Types; mechanisms; changes involved in succession; concept of climax. **G. Ecosystem:** Structure and function; energy flow and mineral cycling (CNP); primary production and decomposition; structure and function of some Indian ecosystems: terrestrial (forest, grassland) and aquatic (fresh water, marine, eustarine). **H. Biogeography:** Major terrestrial biomes; theory of island biogeography; biogeographical zones of India. **I. Applied ecology:** Environmental pollution; global environmental change; biodiversity-status, monitoring and documentation; major drivers of biodiversity change; biodiversity management approaches. **J. Conservation biology:** Principles of conservation, major approaches to management, Indian case studies on conservation/management strategy (Project Tiger, Biosphere reserves).
	Economic Zoology a. Pearl, prawn, carp, vermiculture, apiculture, sericulture, lac culture, and so on. b. The main infectious diseases (AIDS, cholera, tuberculosis, malaria, and filaria), as well as their pathogens, vectors, and preventative measures. c. Diseases affecting cattle and other livestock, along with the helminth pathogens and vectors (ticks, mites, Tabanus, and stromoxys). d. Pests of rice (Sitophilus oryzae), oil seed (Achaeajanata), and sugarcane (Pyrilla perpusilla). e. Animals those are transgenic. f. Gene therapy, human genetic illness, genetic counseling, and medical technologies. g. Biotechnology for forensics
	Evolution and Systematics a. Origin of life theories.: Natural selection, the function of mutation in evolution, evolutionary patterns, molecular drive, mimicry, variety, isolation, and speciation are among the theories of evolution b. Evolution of humans, horses, and elephants based on fossil evidence. c. Law of Hardy-Weinberg. d. Continental drift and dispersal of animals.
3	**UNIT- III**
	Cell Biology a. Cell division (mitosis and meiosis), mitotic spindle and mitotic apparatus, chromosome movement (chromosome type polytene and lampbrush), heterochromatin organization, and cell cycle regulation. Structure and function of the cell and its organelles (nucleus, plasma membrane, mitochondria, Golgi bodies, endoplasmic reticulum, ribosomes, and lysosomes).

	b. Protein foldings and transport; nucleic acid topology; DNA motif; DNA replication; transcription; RNA processing; and translation.
	Genetics Genes, split genes, genetic control, and genetic code are modern concepts. a. The evolution of sex chromosomes, sex determination in humans and Drosophila. b. Mendel's rules of inheritance, multiplication of alleles, recombination, linkage, blood group genetics, pedigree analysis, and hereditary illnesses in humans. c. Mutagenesis and mutations. d. Transgenics, artificial chromosomes as vectors, plasmids, cosmids, recombinant DNA technologies, DNA cloning, and whole animal cloning (principles and procedures). e. Prokaryotes and eukaryotes: gene expression and control. f. Defects in the signaling system, cell death, signal molecules, and their effects.
	Developmental Biology **A. Basic concepts of development:** Potency, commitment, specification, induction, competence, determination and differentiation; morphogenetic gradients; cell fate and cell lineages; stem cells; genomic equivalence and the cytoplasmic determinants; imprinting; mutants and transgenics in analysis of development. **B. Gametogenesis, fertilization and early development:** Production of gametes, cell surface molecules in sperm-egg recognition in animals; embryo sac development and double fertilization in plants; zygote formation, cleavage, blastula formation, embryonic fields, gastrulation and formation of germ layers in animals; embryogenesis. **C. Morphogenesis and organogenesis in animals:** Cell aggregation and differentiation in Dictyostelium; Axes and pattern formation in Drosophila, amphibia and chick; Organogenesis – vulva formation in Caenorhabditis elegans; eye lens induction, limb development and regeneration in vertebrates; differentiation of neurons, post embryonic development-larval formation, metamorphosis; environmental regulation of normal development; sex determination. **D. Programmed cell death, aging and senescence.**
4	**UNIT- IV**
	Biochemistry Molecular taxonomy, biodiversity, cladistics, international code, and zoological nomenclature. a. The composition and functions of proteins, amino acids, lipids, fatty acids, cholesterol, and nucleic acids. bioenergetics. b. The Krebs cycle and glycolysis, glycogenolysis, oxidation and reduction, oxidative phosphorylation, energy release and conservation, ATP, and the structure and function of cyclic AMP. c. The biosynthesis, functioning, and classification of hormones (peptide and steroid hormones). d. Types and modes of action of enzymes. e. Co-enzymes and vitamins. f. Immunity and immunoglobulin.
	Animal Physiology a. **Blood and circulation:** Blood corpuscles, haemopoiesis and formed elements, plasma function, blood volume, blood volume regulation, blood groups, haemoglobin, immunity, haemostasis. b. **Cardiovascular System:** Comparative anatomy of heart structure, myogenic heart, specialized tissue, ECG – its principle and significance, cardiac cycle, heart as a pump, blood pressure, neural and chemical regulation of all above. c. **Respiratory system:** Comparison of respiration in different species, anatomical considerations, transport of gases, exchange of gases, waste elimination, neural and chemical regulation of respiration. d. **Nervous system:** Neurons, action potential, gross neuroanatomy of the brain and spinal cord, central and peripheral nervous system, neural control of muscle tone and posture. e. **Sense organs:** Vision, hearing and tactile response. f. **Excretory system:** Comparative physiology of excretion, kidney, urine formation, urine concentration, waste elimination, micturition, regulation of water balance, blood volume, blood pressure, electrolyte balance, acid-base balance.

	g. **Thermoregulation:** Comfort zone, body temperature – physical, chemical, neural regulation, acclimatization. h. **Stress and adaptation**
	Research Methodology Objectives, Types, Significance , Research process, criteria of good research, Problems encountered by researchers, Research Problem and Research Design, Sampling Design and Methods of Data Collection. Interpretation and Report Writing, Interpretation and Presentation of data, Types of Report, Report Format, Typing instructions and oral Presentation. Ethics in Research: Bioethics: Bioethics and Sub-disciplines in Bioethics, Introduction- Medical Ethics, Animal Ethics, Environmental Ethics, The Idea of Moral Status in Bioethics, Theories in Bioethics: Introduction -Deontological Approaches, Utilitarianism, The Four-Principle Approach, Virtue Ethics, Casuistry, Feminist Bioethics.
	Biostatistics: Experimental Design Measures of central tendency and dispersal; probability distributions (Binomial, Poisson and normal); sampling distribution; difference between parametric and non-parametric statistics; confidence interval; errors; levels of significance; regression and correlation; t-test; analysis of variance; X2 test;; basic introduction to Muetrovariate statistics.
	Introduction to Bioinformatics a) Aim and branches of Bioinformatics, Application of Bioinformatics, Role of internet and www in bioinformatics. b) Basic biomolecular concepts: Protein and amino acid, DNA & RNA, Sequence, structure and function. c) Forms of biological information, Types of Nucleotide Sequence: Genomic DNA, Complementary DNA (cDNA), Recombinant DNA (rDNA), Expressed sequence tags (ESTs), Genomic survey sequences (GSSs). d) DNA sequencing methods: Basic and Automated DNA sequencing, DNA sequencing by capillary array and electrophoresis, Gene expression data. **Bioinformatics data and databases: Sequence Databases** a) Nucleotide sequence Databases: GenBank, EMBL, DDBJ; Protein sequences b) Databases: Swiss-Prot, TrEMBL, UniProt, UniProtKB, UniParc, UniRef, UniMES; c) Sequence motifs Databases: Prosite, ProDom, Pfam, InterPro, Gene Ontology; Sequence file formats: GenBank, FASTA, PIR, ALN/ClustalW2. d) Enzymatic catalysis databases: KEGG ENZYME database; Protein-Protein interaction database: STRING; Chemical Structure databases: Pubchem; Gene Expression databases: GEO, SAGE

दिनांक – २१ जुलै, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग